

Interactive Dashboard & Tableau Storytelling

1. Explain the difference between a Tableau Dashboard and a Tableau Story Point. Give two healthcare-related use cases.

- **Answer:-** The Core Difference

- **Tableau Dashboard:** A single, consolidated view of multiple worksheets (charts, tables, and KPIs) arranged on a single screen. Dashboards are designed to be interactive and exploratory, allowing users to filter, drill down, and discover insights on their own.
- **Tableau Story Point:** A sequence of specific sheets or dashboards arranged in a linear, narrative flow. Each "Story Point" captures a specific state of a visualization or dashboard at a precise moment in time, designed to walk a viewer through a guided narrative or presentation.

Healthcare Use Cases

- **Dashboard Use Case:** *A Hospital Operations Command Center Dashboard.* It displays real-time metrics like current ER waiting times, available ICU beds, and scheduled surgeries. Shift managers use it continuously throughout the day to interactively monitor and reallocate staff.
- **Story Point Use Case:** *An Annual Public Health Outcomes Report.* A public health official creates a 5-step story tracking a recent outbreak: Point 1 shows patient distribution geographically, Point 2 isolates the timeline, Point 3 shows treatment costs, and Point 4 highlights the final recovery rates to pitch for a budget increase.

2. What are Actions in Tableau? Explain how Filter Action can help hospital management analyze patient data.

- **Answer:-** Actions are interactive user triggers established within a Tableau workflow that allow you to inject contextual logic between worksheets. They enable selections made via mouse-click,

hover, or menu tooltips on one chart to seamlessly alter data presentation across other charts.

How Filter Actions Benefit Hospital Management Data Analysis

- A Filter Action explicitly uses a chosen target data mark from a source sheet to instantaneously filter all remaining records shown across other charts on a unified dashboard.
- Example: If hospital managers click on the "Critical" status slice inside a *Patient Outcome Analysis Pie Chart*, an active Filter Action will dynamically update the adjacent *Department Cost Bar Chart* and *Top Doctors Chart* to display metrics *solely* for those critical patients. This lets management rapidly isolate operational risks, bottlenecked departments, or high-cost patient segments without sorting through massive raw spreadsheets manually

3. Patient Distribution

- **Create a Bar Chart showing:**
- **Total Patients by Hospital Region.**
- **Add this as a Story Point with proper title**

- **Answer:-** Patient Distribution (Bar Chart)

1. Build: Drag **Hospital Region** from the Data pane and drop it onto the Columns shelf.
2. Measure: Drag **Patient ID** onto the Rows shelf. Right-click the **Patient ID** pill, hover over Measure, and select Count.
3. Sort: Click the Sort Descending button (icon with bars and a down arrow) on the top toolbar.
4. Story: Click the New Story tab at the bottom of the screen. Drag your bar chart sheet from the left pane into the center workspace. Double-click the caption box at the top and type Patient Distribution by Hospital Region.

[[1](#), [2](#), [3](#), [4](#), [5](#)]

4. Department Performance

- Create a visualization showing:
- Treatment Cost by Department.
- Add labels and customize colors.

- **Answer:-** Department Performance (Visual Chart)

1. Build: Drag **Department** to the Columns shelf and **Treatment Cost** to the Rows shelf.
2. Label: Drag **Treatment Cost** from the Data pane and drop it directly onto the Label card inside the Marks card.
3. Color: Drag **Department** from the Data pane and drop it onto the Color card.
4. Palette: Click the Color card, click Edit Colors, select a palette from the drop-down menu, and click OK.

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5. Treatment Trend Over Time

- Create a Line Chart:
- Treatment Cost vs Admission Date.
- Filter the data only from March to September.

- **Answer:-** Treatment Trend Over Time (Line Chart)

1. Date Config: Drag **Admission Date** to the Columns shelf. Right-click the **Admission Date** pill on the shelf and select the second "Month" option in the menu (the continuous green date part that shows an example like *May 2015*). [[1](#), [2](#)]
2. Build: Drag **Treatment Cost** to the Rows shelf.
3. Filter: Drag **Admission Date** from the Data pane and drop it onto the Filters shelf. [[1](#), [2](#), [3](#)]
4. Select Months: In the pop-up window, select Months and click Next. Check the boxes for March, April, May, June, July, August, and September, then click OK.

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6. Outcome Analysis

- Create a Pie Chart:

- **Patient Outcome Distribution (Recovered / Critical / Under Treatment).**

- **Answer:-** Outcome Analysis (Pie Chart)

- Chart Type: On the Marks card, change the drop-down menu from Automatic to Pie.
- Color: Drag Patient Outcome onto the Color card.
- Angle: Drag Patient ID onto the Angle card. Right-click it and change the measure to Count.
- Label: Drag Patient ID onto the Label card. Right-click it and change the measure to Count.

7. Doctor Performance & Patient KPI Analysis

Task Part A

- **Doctor Performance Create a Ranked Bar Chart showing:**

- **Top Doctors by Total Treatment Cost handled**

- **Answer:-** Part A: Doctor Performance (Ranked Bar Chart)

- Columns: Drag Treatment Cost to the Columns shelf.
- Rows: Drag Doctor Name to the Rows shelf.
- Sorting: Click the Sort Descending icon on the top toolbar to rank the doctors.
- Top 10 Filter: Drag Doctor Name to the Filters shelf. Go to the Top tab, click By field, set it to Top 10 by Treatment Cost (Sum), and click OK.

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Task Part B

- **KPI Creation**

- **Create a KPI visualization showing:**

- **Total Patients Treated**

- **Answer:-** KPI Creation

- Text: Open a brand-new worksheet. Drag Patient ID directly onto the Text card on the Marks card. Right-click it and change the measure to Count. [[1](#)]
- Formatting: Click the Text card, then click the three dots (...) next to Edit Text. Highlight the text, change the size to 24pt, make it Bold, and format the label to read:
[COUNT(Patient ID)] Total Patients Treated

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8. Interactive Dashboard

- **Build a Dashboard containing:**
- **Region Chart**
- **Department Cost Chart**
- **Outcome Chart**
- **Add Filters:**
- **Hospital Region**
- **Department**

- **Answer:-** Step 1: Create and Size the Dashboard

1. Go to the bottom tabs of your Tableau workspace and click the New Dashboard icon (the window pane symbol with a plus sign).
2. Go to the left-hand Size pane.
3. Click the dropdown menu and change it from Range to Fixed Size (e.g., Generic Desktop or Laptop) so your grid elements do not shift around. [[1](#), [2](#), [3](#), [4](#), [5](#)]

Step 2: Drag and Arrange Your Worksheets

1. Find your sheets listed under the Sheets section on the left sidebar.
2. Drag the Region Chart (Patient Distribution) and drop it onto the blank canvas.
3. Drag the Department Cost Chart (Department Performance) and hover near the right edge of the screen until a grey shadow box appears, then drop it to snap it side-by-side with the region chart.

4. Drag the Outcome Chart (Pie Chart) and drop it along the bottom edge of the layout to create your completed grid. [[1](#), [2](#)]

Step 3: Configure Global Sheet Filters

1. Click once inside your Region filter box (or Department filter box) on the right side of the dashboard to select it.
2. Click the small down arrow (More Options) on its right-hand border frame.
3. Hover your mouse over Apply to Worksheets in the context menu.
4. Click All Using This Data Source.
5. Repeat these exact steps for your Department filter card to sync your filters across every single chart. [[1](#)]

Step 4: Activate Cross-Filtering (Use as Filter) [[1](#)]

1. Click once inside the Region Chart layout zone to bring up its grey outer border line.
2. Click the tiny funnel icon ("Use as Filter") located in the upper-right control frame of that chart.
3. Click inside the Department Cost Chart zone and click its funnel icon.
4. Click inside the Outcome Chart zone and click its funnel icon.

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9. Storytelling Flow

Create a Tableau Story with minimum 4 Story Points:

- Patient Distribution
- Department Cost Analysis
- Outcome Analysis
- Doctor Performance

- **Answer:-** Step 1: Create a New Story Workspace

1. Go to the bottom tab bar of your Tableau window.

2. Click the New Story icon (the icon looks like a book or a blank page with a bookmark symbol).
3. In the left-hand sidebar, under Story Size, click the dropdown and choose a layout size (like Generic Desktop) to match your dashboard presentation scale.

Step 2: Build Chronological Presentation Tabs

Story Point 1: Patient Distribution

1. From the left sidebar list of sheets, drag your Patient Distribution bar chart and drop it into the center canvas area.
2. Double-click the grey navigator text box at the top that says *"Write the story point caption here"*.
3. Type: "1. Geographic Patient Footprint"

Story Point 2: Department Cost Analysis

1. Look at the left sidebar menu and click Blank under the *New Story Point* section to create your second slide.
2. Drag your Department Cost Chart (or the complete Interactive Dashboard you compiled) into the center canvas.
3. Double-click the grey caption text box at the top.
4. Type: "2. Financial Expenditures by Specialization"

Story Point 3: Outcome Analysis

1. Click Blank under the *New Story Point* section on the left sidebar to add a third slide.
2. Drag your Outcome Analysis pie chart from the sidebar and drop it onto the canvas.
3. Double-click the grey caption text box at the top.
4. Type: "3. Summary of Treatment Recovery & Care Success Rates"

Story Point 4: Doctor Performance

1. Click Blank under the *New Story Point* section one last time to create your fourth slide.
2. Drag your Doctor Performance ranked bar chart into the center canvas view.
3. Double-click the grey caption text box at the top.
4. Type: "4. Doctor Workload Handling & Top Providers"

10. Based on your visuals, answer:

- **Which department generates the highest treatment cost?**

- **Answer:-** Evaluate your Department Performance Chart. The single longest bar (accompanied by the largest aggregated data label value) points directly to your answer. Specialized clinical sectors such as *Cardiology* or *Surgery* frequently take the lead due to extensive operating room resources or surgical implant costs.

- **Which region has more critical patients?**

- **Answer:-** Navigate to your interactive dashboard. Click directly on the "Critical" slice of your Outcome Pie Chart. Thanks to the custom Filter Action you built, the Region Chart will instantly update to show *only* the geographic location of critical patients. The tallest remaining regional bar identifies the highest volume area

- **What business decisions can hospital management take from these insights?**

- **Answer:-** Targeted Allocation of Emergency Care Resources: Move additional clinical staff, critical care equipment, and trauma-capable ambulance vehicles into the specific region highlighted with the greatest volume of critical outcomes.
- Clinical Path Optimization & Standardizations: Audit the operations of the highest-costing clinical department. By

matching those records against your ranked Doctor Performance chart, you can determine if a tiny subset of staff is driving up treatment averages through excessive lengths of stay or non-standard protocols.

- Seasonal Staffing Adjustments: Review the trajectory of the March–September Treatment Trend over time chart. If consistent seasonal spikes emerge (e.g., higher admissions in late summer months), manage operational staffing rotas to handle busy intake patterns seamlessly.

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